SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I H. Divyaa Dhaarini (192426074) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: H. Divyaa dhaarini

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	1	1	1	1	1
Debugging	25%	1	1	1	1	1
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	1	1

25/30

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
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Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
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- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

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Analysis	20%					
Code Quality	15%					

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Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Answer (8 Marks)

Given: Workstation IP 192.168.10.65, mask 255.255.255.192 (/26), default gateway 192.168.10.1. It must communicate with hosts in 192.168.10.0/26.

Method: /26 means host bits = 6 and block size = 2^6 = 64, so the subnets begin at .0, .64, .128 and .192. Network address = IP AND mask; broadcast = network + 63. A host and its default gateway must be in the same subnet.

Step 1: Subnet of the configured IP address

65 = 01000001; 192 = 11000000; AND = 01000000 = 64

The workstation belongs to subnet 192.168.10.64/26, not to 192.168.10.0/26.

Step 2: Network, broadcast and host range

Parameter	Workstation subnet	Intended subnet / gateway subnet
Network address	192.168.10.64/26	192.168.10.0/26
Valid host range	192.168.10.65 – 192.168.10.126	192.168.10.1 – 192.168.10.62
Broadcast address	192.168.10.127	192.168.10.63
Usable hosts	62	62

Step 3: Validity check

IP 192.168.10.65: valid host address (first usable host of 192.168.10.64/26)

Gateway 192.168.10.1: valid host address, but belongs to 192.168.10.0/26

Gateway check: 1 AND 192.168.10.64 mask gives network .0, which is not .64

Each address is valid on its own, but the pair is invalid: the gateway is not in the workstation's subnet.

Step 4: Configuration errors

Error 1: The workstation IP (192.168.10.65) is in the wrong subnet. It lies in 192.168.10.64/26, so it is not in the 192.168.10.0/26 subnet of the systems it must reach.

Error 2: The default gateway 192.168.10.1 is outside the workstation's subnet, so the host cannot ARP for it and treats it as unreachable. Traffic to other subnets cannot leave.

Step 5: Optimized configuration

Setting	Corrected value
IP address	192.168.10.2 (any free host between .2 and .62)
Subnet mask	255.255.255.192 (/26)
Default gateway	192.168.10.1 (router interface, first usable address)

Now the host and the gateway are both in 192.168.10.0/26 (2 AND 192 = 0). Verification in Cisco Packet Tracer: ping 192.168.10.1 and a neighbour in the subnet, then check ipconfig on the PC.

Step 6: Usable hosts after correction. 2^6 − 2 = 62 usable addresses in 192.168.10.0/26 (192.168.10.1 – 192.168.10.62). One is used by the gateway, so 61 remain for workstations.

Interpretation: The fault is a subnet mismatch, not a bad address. Moving the host into the correct /26 and keeping the gateway in the same subnet restores communication.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer (8 Marks)

Given: Device A: 2001:db8::ff00:42:8329/64 and Device B: 2001:db8:0:0:1::1/64.

Step 1: Expansion to complete notation

Each group is padded to 4 hex digits and '::' is expanded with enough zero groups to make 8 groups. Device A has 5 written groups, so '::' stands for 3 zero groups. Device B has 6 written groups, so '::' stands for 2 zero groups.

Device A: 2001:db8::ff00:42:8329 → 2001:0db8:0000:0000:0000:ff00:0042:8329
Device B: 2001:db8:0:0:1::1 → 2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network prefix (first 64 bits = first four groups)

Device	First four groups	Network prefix	Interface ID
A	2001:0db8:0000:0000	2001:db8::/64	0000:ff00:0042:8329
B	2001:0db8:0000:0000	2001:db8::/64	0001:0000:0000:0001

Step 3: Same subnet?

Yes. Both devices have the prefix 2001:0db8:0000:0000 with /64, so they are in the same subnet 2001:db8::/64. The '1' in the fifth group of Device B is part of its interface ID, not of the network prefix.

Step 4: Mismatch analysis

There is no prefix mismatch in the configuration as given: prefix length and network bits are identical. The apparent problem is that Device B's address looks as if '1' were a subnet ID (2001:db8:0:0:1::/80 style), which is a misreading. Since addressing is correct, the failure must come from outside the address: for example the devices are not on the same VLAN or link, an interface is down or IPv6 is not enabled, an ACL or firewall blocks ICMPv6 / Neighbor Discovery, or the address was mistyped (for example 2001:db8:0:1::1/64 would be a different subnet, 2001:db8:0:1::/64). Checking with ping and the Wireshark ICMPv6 neighbour solicitation/advertisement packets would confirm which.

Step 5: Optimized configuration

Device	Optimized address	Prefix
Device A	2001:db8::10/64	2001:db8::/64
Device B	2001:db8::20/64	2001:db8::/64

Short, regular interface IDs make the plan easy to read and to document, and keep both devices in one /64 on the same link (enable IPv6 with ipv6 address 2001:db8::10/64 and ipv6 enable on the interfaces).

Step 6: Host addresses in the subnet. Host bits = 128 − 64 = 64, so the subnet has 2^64 = 18,446,744,073,709,551,616 addresses. IPv6 has no broadcast address, and if the subnet-router anycast address (2001:db8::) is excluded, 2^64 − 1 are assignable.

Interpretation: Both devices are correctly in 2001:db8::/64, so the cause of the failure lies in the link or device configuration, not in the prefix.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Answer (8 Marks)

Assumption: The question does not list the departments or their host needs, so the following are assumed (replace them with your own figures if given). The existing plan uses one /26, one /27 and one /28 in order: Department A (/26) needs 20 hosts, Department B (/27) needs 45 hosts, and Department C (/28) needs 25 hosts.

Step 1: Existing allocation of 192.168.1.0/24

Dept.	Subnet	Host range	Broadcast	Usable	Needed	Unused / shortage
A	192.168.1.0/26	192.168.1.1 – .62	192.168.1.63	62	20	42 unused
B	192.168.1.64/27	192.168.1.65 – .94	192.168.1.95	30	45	15 short
C	192.168.1.96/28	192.168.1.97 – .110	192.168.1.111	14	25	11 short

Usable hosts = 2^(32 − prefix) − 2, giving 62, 30 and 14. Addresses 192.168.1.112 – 192.168.1.255 (144 addresses) are not allocated to any department.

Step 2: Inefficient allocations and mistakes

Mistake 1: The subnet sizes were not matched to demand. The largest block (/26) went to the department with the smallest need (A), wasting 42 addresses (68% of the block).

Mistake 2: Department B (needs 45) was given a /27 with only 30 usable hosts, and Department C (needs 25) a /28 with only 14, so both run out of addresses.

Mistake 3: A fixed-size approach was used instead of VLSM, and the remaining 144 addresses sit idle while departments have shortages.

Step 3: Optimized VLSM scheme

Sort the departments by size (largest first) and pick the smallest block that fits: 45 hosts needs a /26 (62), 25 hosts needs a /27 (30), and 20 hosts needs a /27 (30).

Dept.	Needed	Subnet	Host range	Broadcast	Usable	Spare
B	45	192.168.1.0/26	192.168.1.1 – .62	192.168.1.63	62	17
C	25	192.168.1.64/27	192.168.1.65 – .94	192.168.1.95	30	5
A	20	192.168.1.96/27	192.168.1.97 – .126	192.168.1.127	30	10

Result: The three subnets use 64 + 32 + 32 = 128 addresses. The block 192.168.1.128/25 (192.168.1.128 – 192.168.1.255, 128 addresses) is entirely free and can be used for new departments, or split into /30 subnets for router links.

Interpretation: VLSM removes the shortages in B and C and cuts the waste in A. In the existing plan Department A wastes 42 addresses while B and C are short by 15 and 11; in the optimized plan every department fits with 5 to 17 spare addresses, and half of the /24 (128 addresses) stays free for growth.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
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- Verifying whether both devices belong to the same subnet.
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- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answer (8 Marks)

Note: Question 4 is identical to Question 2 (same devices, same tasks), so the same analysis and result apply.

Given: Device A: 2001:db8::ff00:42:8329/64 and Device B: 2001:db8:0:0:1::1/64.

Step 1: Expansion to complete notation

Each group is padded to 4 hex digits and '::' is expanded with enough zero groups to make 8 groups. Device A has 5 written groups, so '::' stands for 3 zero groups. Device B has 6 written groups, so '::' stands for 2 zero groups.

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Step 2: Network prefix (first 64 bits = first four groups)

Device	First four groups	Network prefix	Interface ID
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Yes. Both devices have the prefix 2001:0db8:0000:0000 with /64, so they are in the same subnet 2001:db8::/64. The '1' in the fifth group of Device B is part of its interface ID, not of the network prefix.

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Short, regular interface IDs make the plan easy to read and to document, and keep both devices in one /64 on the same link (enable IPv6 with ipv6 address 2001:db8::10/64 and ipv6 enable on the interfaces).

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Interpretation: Both devices are correctly in 2001:db8::/64, so the cause of the failure lies in the link or device configuration, not in the prefix.

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Answer (8 Marks)

Given: Routing table: 192.168.1.0/24 via 10.0.0.2; 192.168.2.0/24 via 10.0.0.3; default route via 10.0.0.1. Destination: 192.168.2.100.

Step 1: Longest-prefix match

Table entry	**Mask**	**Destination AND mask**	**Match?**
192.168.1.0/24	255.255.255.0	192.168.2.0 ≠ 192.168.1.0	No
192.168.2.0/24	255.255.255.0	192.168.2.0 = 192.168.2.0	Yes, prefix /24
Default (0.0.0.0/0)	0.0.0.0	0.0.0.0 = 0.0.0.0	Yes, prefix /0

Two entries match; the /24 prefix is longer than /0, so the router selects 192.168.2.0/24.

Step 2: Next hop selected. 10.0.0.3.

Step 3: Does the destination belong to the intended subnet? Yes. 192.168.2.0/24 has hosts 192.168.2.1 – 192.168.2.254 and broadcast 192.168.2.255, so 192.168.2.100 is a valid host in it.

Step 4: Configuration errors

According to the table the forwarding decision is correct: the packet matches the right entry and goes to 10.0.0.3. So the likely errors are in the details of that entry and its surroundings:

1. 10.0.0.3 may be the wrong or unreachable next hop (down, wrong IP, or no ARP entry); the entries for 192.168.1.0 and 192.168.2.0 may have been swapped.
2. 10.0.0.3 may lack a route to 192.168.2.0/24 or a return route to the sender.
3. The static routes name only a next-hop IP and no exit interface, and there is no metric or backup path.
4. An ACL or firewall on the path may drop traffic to 192.168.2.0/24.

Verification (Packet Tracer / Wireshark): show ip route 192.168.2.100, ping 10.0.0.3, show ip arp and traceroute 192.168.2.100.

Step 5: Optimized routing table

Destination network	Next hop / interface	Purpose
192.168.1.0/24	10.0.0.2 via Gi0/0	Specific route to Network 1
192.168.2.0/24	10.0.0.3 via Gi0/0 (verified reachable)	Specific route to Network 2
0.0.0.0/0	10.0.0.1	Default route for all other traffic

ip route 192.168.1.0 255.255.255.0 GigabitEthernet0/0 10.0.0.2

ip route 192.168.2.0 255.255.255.0 GigabitEthernet0/0 10.0.0.3

ip route 0.0.0.0 0.0.0.0 10.0.0.1

Giving both exit interface and next hop avoids recursive lookups and ARP problems. A floating backup route with a higher administrative distance, or a dynamic protocol such as OSPF, can add redundancy as the network grows. The two /24 networks cannot be summarized, because they use different next hops.

Step 6: Route used when no match exists. The default route 0.0.0.0/0 via 10.0.0.1 (for example for 192.168.3.5 or 8.8.8.8).

Interpretation: Longest-prefix matching sends 192.168.2.100 to 10.0.0.3, the intended next hop. The fault must therefore be fixed at the next-hop router or link rather than in the choice of entry.

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